The critical scaling of the uniform doubling condition: smooth bodies versus truncations

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Abstract

The uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] is a property of a training set $T \subset \mathbb{R}^n$ guaranteeing stability of accuracy during the training of neural networks with piecewise linear activations: for every truncated slab and every width on a geometric grid between a scaling factor ℓ and a scale parameter β, widening the slab by the factor κ must multiply its empirical mass by $1+\sigma$ unless the mass already exceeds δ. In the companion paper [*The uniform doubling condition holds with high probability for independent samples*] we proved that N i.i.d. samples with density bounded above and below on a convex body K satisfy the condition with probability $1-1/N$, with N-independent $(\kappa,\delta,\sigma,\beta)$ and scaling factor $\ell_N = O\big((\log N/N)^{1/n}\big)$, and that some power-law decay of ℓ_N is necessary: on the ball, the condition fails with probability bounded below unless $\ell \geq c\,N^{-2/(n+1)}$. This left a gap between the exponents $1/n$ and $2/(n+1)$. Here we close the gap and reveal a dichotomy. For *pure slabs* (no truncations) over a body with an interior rolling ball, a depth-integral estimate $\mathrm{vol}\big(K \cap \{|\langle y-x,u\rangle| \leq L\}\big) \geq c\,L^{(n+1)/2}$, valid uniformly in $x \in K$ and in the direction u, improves the scaling factor to $\ell_N = O\big((\log N/N)^{2/(n+1)}\big)$: the critical exponent for smooth bodies is exactly $2/(n+1)$, up to logarithms. For *truncated* slabs the improvement is impossible: already one fixed, uniformly thick truncation of the cube creates, with probability bounded below, a diagonal-level gap beyond the cornermost sample point which forces $\ell \geq c\,N^{-1/n}$. Thus the truncation number, not the smoothness of the support, dictates the critical scaling: $2/(n+1)$ without truncations, $1/n$ with them.

1 Introduction

Let $K \subset \mathbb{R}^n$ be a convex body, let μ be a probability measure with density ρ on K satisfying

$$0 < \rho_{\min} \leq \rho \leq \rho_{\max} < \infty \ \text{ on } K, \qquad \widetilde{R} := \rho_{\max}/\rho_{\min}, \tag{1}$$

and let $T = \{X_1, \dots, X_N\}$ be i.i.d. samples from μ with empirical measure $\bar{\mu}_N$. For a unit vector u, a center $s \in \mathbb{R}^n$, a width $w > 0$ and a truncation set $h = \{(v_i, t_i)\}_{i \leq k}$, the truncated slab is

$$S_w \ = \ \Big\{x : \ |\langle x-s,u\rangle| \leq \tfrac{w}{2}\Big\} \cap H, \qquad H = \{x : \ \langle v_i, x\rangle \leq t_i, \ i \leq k\},$$

with truncation region $G := H \cap K$. The *uniform doubling condition* (UDC) of [2], with constants $(\kappa,\delta,\sigma,\beta,\ell)$ and truncation family $\mathcal{L}$, requires

$$\bar{\mu}_N(S_{\ell\kappa^{i+1}}) \ \geq \ \min\big\{\delta, \ (1+\sigma)\,\bar{\mu}_N(S_{\ell\kappa^i})\big\}$$

for every truncated slab generated by $\mathcal{L}$ and every $i \in \mathbb{N}_0$ with $\ell\kappa^i < \beta$. The condition is the hypothesis of the stability-of-accuracy theorems of [1, 2]. In the companion paper [3] we proved:

Theorem 1.1 ([3])**.** *Under (1), for every k and $r_0 > 0$ there are N-independent constants $\kappa = 1 + 150\widetilde{R}$, $\sigma = 1$, $\delta = \frac{3\rho_{\min}\omega_n r_0^n}{32\widetilde{R}}$, $\beta = \operatorname{diam} K$, such that with probability at least $1 - \frac{1}{N}$ the sample satisfies the UDC, simultaneously for all truncation families of size k whose nonempty regions G contain a ball of radius r_0, with scaling factor*

$$\ell_N \;=\; C_0\Big(\frac{\widetilde{R}}{\rho_{\min}}\Big(\frac{\operatorname{diam} K}{r_0}\Big)^{n-1}\frac{\log N}{N}\Big)^{1/n}.$$

Moreover, for the uniform measure on the unit ball and untruncated slabs, with probability bounded below by a constant the condition fails for every $\ell \le c\,N^{-2/(n+1)}$ (for any fixed $\kappa > 1$, $\sigma > 0$, and $\delta \ge 2/N$).

The exponents do not match: $1/n < \frac{2}{n+1}$ for $n \ge 2$, i.e. the sufficient scaling factor is larger than the necessary one. This paper determines the critical exponent, with an answer that depends not on the regularity of K but on the presence of truncations.

Theorem 1.2 (Smooth bodies, pure slabs)**.** *Let K have the interior rolling ball property at radius $\rho_K > 0$: for every boundary point p there is a ball $B \subseteq K$ of radius ρ_K with $p \in \partial B$ (true, e.g., for any K with C^2 boundary and strictly positive principal curvatures; for the unit ball, $\rho_K = 1$). Assume (1) and consider the UDC for* untruncated *slabs ($k = 0$, $G = K$). Then with probability at least $1 - \frac{1}{N}$ the sample satisfies the UDC with the constants $(\kappa, \sigma, \delta, \beta)$ of Theorem 1.1 (with $r_0 = \rho_K$) and scaling factor*

$$\ell_N \;=\; C_1\Big(\frac{\log N}{N}\Big)^{2/(n+1)}, \qquad C_1 = C_1\big(n, \widetilde{R}, \rho_{\min}, \rho_K\big).$$

Together with the lower bound of Theorem 1.1, the critical exponent for bodies with a rolling ball and pure slabs is $\frac{2}{n+1}$, up to the logarithmic factor.

Theorem 1.3 (Truncations shift the exponent)**.** *Let $K = [0,1]^n$ with $\mu = \mathrm{Leb}$, $n \ge 1$, and fix any $\kappa > 1$, $\sigma > 0$, $\beta > 0$ and $\delta \ge \frac{3}{N}$. Consider the single truncation set $h^* = \{(-\mathbf{1}/\sqrt{n},\ -(n - \frac{1}{2})/\sqrt{n})\}$, i.e.*

$$G^* \;=\; \Big\{x \in [0,1]^n :\ \textstyle\sum_j x_j \ge n - \frac{1}{2}\Big\},$$

a corner cap containing a ball of radius $r_0(n) = \frac{1}{4n}$. There are constants $c_2(n,\kappa), c_3(n,\kappa) > 0$ such that for all large N, with probability at least c_3, the doubling inequality fails for a truncated slab generated by h^* at every *grid scale $\le c_2 N^{-1/n}$; hence the UDC with truncation number $k \ge 1$ forces*

$$\ell \;\ge\; c_2\,N^{-1/n}.$$

Consequently the scaling factor of Theorem 1.1 is sharp up to logarithms in the truncated setting — even for a single, fixed, uniformly thick truncation — and the critical exponent is $\frac{1}{n}$ with truncations versus $\frac{2}{n+1}$ without.

The mechanism is geometric. The empirical argument of [3] requires that any (truncated) slab of width $w \ge \ell_N$ containing a sample point have, after widening by κ, population mass of order at least $\log N/N$, so that the Vapnik–Chervonenkis relative deviation bounds take over.

Without truncations, the widened slab around *any* point of a rolling-ball body captures at least the volume of a spherical-cap stack, $\asymp L^{(n+1)/2}$ in the half-width L (Lemma 3.1); equating with $\log N/N$ gives the exponent $\frac{2}{n+1}$. With truncations, the slab may live inside a corner region of G whose level sections degenerate polynomially, where the captured mass is only $\asymp L^n$; the matching lower-bound construction places the slab at the cornermost sample point, beyond which a level gap of relative size κ^2 occurs with probability bounded below, stalling the empirical mass at $1/N$ through an entire grid step.

In the regime relevant to deep networks — where the truncations are the activation cells, $k \geq 1$ — the exponent $\frac{1}{n}$ of [3] is therefore correct; for the pure-slab doubling conditions of Berlyand–Jabin–Safsten type [1] on smooth data supports, the faster scaling $\frac{2}{n+1}$ is available.

2 Ingredients from the companion paper

We reuse three components of [3].

Proposition 2.1 (Population dichotomy, [3])**.** *Let $\kappa \geq 4$ and let S_w be a truncated slab whose region G contains a ball of radius r_0. Then for every $w > 0$,*

$$\mu(S_{\kappa w}) \;\geq\; \min\Big\{\delta_0,\; \Big(1 + \tfrac{\kappa-1}{15\widetilde{R}}\Big)\mu(S_w)\Big\}, \qquad \delta_0 = \frac{3\rho_{\min}\omega_n r_0^n}{16\widetilde{R}}.$$

Lemma 2.2 (VC relative deviations, [3])**.** *Let $d = d_{n,k}$ be the VC bound for intersections of $k+2$ half-spaces and $u_N = \frac{4}{N}\big(d\ln\frac{2eN}{d} + \ln(8N)\big)$. With probability at least $1 - \frac{1}{N}$, simultaneously for all truncated slabs C: if $\mu(C) \geq 16u_N$ then $\bar{\mu}_N(C) \geq \frac{3}{4}\mu(C)$; and if $\bar{\mu}_N(C) \geq 4u_N$ then $\mu(C) \geq \frac{1}{2}\bar{\mu}_N(C)$.*

Proposition 2.3 (Proof skeleton, [3])**.** *Let N be large enough that $32u_N \leq 2\delta_0$. On the event of Lemma 2.2, suppose that a scale w has the property that every truncated slab S of width w (region $G \supseteq$ some r_0-ball) satisfies*

$$\bar{\mu}_N(S_w) > 0 \implies \mu(S_{\kappa w}) \;\geq\; 32\,u_N. \tag{2}$$

Then the doubling inequality $\bar{\mu}_N(S_{\kappa w}) \geq \min\{\delta, 2\bar{\mu}_N(S_w)\}$, with $\delta = \delta_0/2$ and $\kappa = 1 + 150\widetilde{R}$, holds at scale w for all such slabs.

Indeed, inspection of the proof of the main theorem of [3] shows that the sample point $x \in S_w \cap G$ is used only to produce the mass bound (2) (there, through a cone with apex x and base an r_0-ball, giving mass $\gtrsim w^n$); every subsequent step — the case analysis combining Proposition 2.1 with Lemma 2.2 — consumes (2) alone. The scaling factor ℓ_N may therefore be taken as any w_0 such that (2) holds for all $w \in [w_0, \beta)$. Improving the geometric lower bound improves the exponent, and this is all we shall do.

3 Pure slabs over rolling-ball bodies

Lemma 3.1 (Depth-integral slab-volume bound)**.** *Let K have the interior rolling ball property at radius ρ_K, and let $n \geq 2$. Then for every $x \in K$, every unit vector u, and every $0 < L \leq \rho_K$,*

$$\mathrm{vol}\Big(K \cap \big\{y :\; |\langle y - x, u\rangle| \leq L\big\}\Big) \;\geq\; a_n\,\rho_K^{(n-1)/2}\,L^{(n+1)/2}, \qquad a_n := \frac{\omega_{n-1}}{2^n},$$

where ω_{n-1} is the volume of the unit ball of $\mathbb{R}^{n-1}$. (For $n = 1$ the left-hand side is at least L, and the statement holds trivially with $a_1 = 1$.)

Proof. Write $\rho := \rho_K$, $\tau(y) := \langle y, u\rangle$, $\tau_{\max} := \max_K \tau$, $\tau_{\min} := \min_K \tau$, and let $A(\tau) := \mathrm{vol}_{n-1}\big(K \cap \{\tau(y) = \tau\}\big)$, so that the slab volume equals $\int_{\tau(x)-L}^{\tau(x)+L} A(\tau)\, d\tau$. Note that K contains a ball of radius ρ, so $\tau_{\max} - \tau_{\min} \geq 2\rho \geq 2L$. By the Brunn–Minkowski inequality, $A^{1/(n-1)}$ is concave on $[\tau_{\min}, \tau_{\max}]$; in particular A is *unimodal*, so on any subinterval A is bounded below by the smaller of its endpoint values.

Step 1: an anchor level near each extreme. Let $p \in \partial K$ attain $\tau_{\max}$ and let $B = B(c, \rho) \subseteq K$ be a rolling ball with $p \in \partial B$; set $g := \tau(p) - \tau(c) \in [0, \rho]$ (nonnegative since $\tau(c) \leq \tau_{\max}$, and at most $|p - c| = \rho$). The section of B at the level $\tau_{\max} - h$ is an $(n-1)$-ball of radius $r(h)$ with $r(h)^2 = \rho^2 - (g-h)^2$. For $0 \leq h \leq g$,

$$r(h)^2 = (\rho - g + h)(\rho + g - h) \;\geq\; h \cdot \rho,$$

since $\rho - g + h \geq h$ and $\rho + g - h \geq \rho$. Hence

$$A(\tau_{\max} - h) \;\geq\; \omega_{n-1}(\rho h)^{(n-1)/2} \qquad (0 \leq h \leq g), \tag{3}$$

and at $h = g$ the section of B is its full equator: $A(\tau_{\max} - g) \geq \omega_{n-1}\rho^{n-1}$.

Define the *anchor depth* $a := \min(g,\, L/4)$ and the value $A^* := \omega_{n-1}(\rho L/4)^{(n-1)/2}$. Then

$$A(\tau_{\max} - a) \;\geq\; A^* : \tag{4}$$

if $g \geq L/4$ then $a = L/4 \leq g$ and (3) gives $A(\tau_{\max} - a) \geq \omega_{n-1}(\rho L/4)^{(n-1)/2} = A^*$; if $g < L/4$ then $a = g$ and the equator bound gives $A(\tau_{\max} - a) \geq \omega_{n-1}\rho^{n-1} \geq A^*$, using $\rho \geq L/4$. Symmetrically, with g' the offset of a rolling ball at a minimizer of τ, the anchor depth $a' := \min(g', L/4)$ satisfies $A(\tau_{\min} + a') \geq A^*$. Note $a, a' \leq L/4$.

Step 2: the integral. By (4) and unimodality,

$$A(\tau) \;\geq\; A^* \qquad \text{for all } \tau \in M := [\tau_{\min} + a',\; \tau_{\max} - a].$$

The complement of M in $[\tau_{\min}, \tau_{\max}]$ has total length $a + a' \leq L/2$. The slab's level interval $[\tau(x) - L, \tau(x) + L]$ intersects $[\tau_{\min}, \tau_{\max}]$ in an interval of length at least L: indeed $\tau(x)$ lies in the range, and the range has length at least $2\rho \geq 2L$, so at least one of the two sides of $\tau(x)$ contains a full segment of length L inside the range. Consequently the slab meets M in a set of length at least $L - \frac{L}{2} = \frac{L}{2}$, and

$$\int_{\tau(x)-L}^{\tau(x)+L} A \;\geq\; \frac{L}{2}\, A^* \;=\; \frac{\omega_{n-1}}{2 \cdot 4^{(n-1)/2}}\, \rho^{(n-1)/2} L^{(n+1)/2} \;=\; \frac{\omega_{n-1}}{2^n}\, \rho^{(n-1)/2} L^{(n+1)/2}.$$

□

Proof of Theorem 1.2. Take $r_0 = \rho_K$ (the rolling ball is contained in K, so the inradius is at least ρ_K, and $G = K$ trivially contains an r_0-ball). Set

$$\ell_N := \frac{2}{\kappa - 1}\Big(\frac{32\, u_N}{\rho_{\min}\, a_n\, \rho_K^{(n-1)/2}}\Big)^{2/(n+1)},$$

with u_N from Lemma 2.2 for $k = 0$, and let N be large enough that $32u_N \leq \min\{2\delta_0,\; \rho_{\min} a_n \rho_K^n\}$ and $\frac{(\kappa-1)\ell_N}{2} \leq \rho_K$. By Proposition 2.3 it suffices to verify (2) for every $w \geq \ell_N$ and every slab S_w ($k = 0$). Let $x \in S_w \cap T$ be a sample point (so $x \in K$ a.s.), and put $L := \frac{(\kappa-1)w}{2}$. Since

$|\langle x-s,u\rangle| \le \frac{w}{2}$, the slab $\{y : |\langle y-x,u\rangle| \le L\}$ is contained in $S'_{\kappa w}(u,s)$, so by Lemma 3.1 (applied with half-width $\min(L,\rho_K)$),

$$\mu(S_{\kappa w}) \;\ge\; \rho_{\min}\, a_n\, \rho_K^{(n-1)/2}\, \min(L,\rho_K)^{(n+1)/2}.$$

If $L \le \rho_K$, the right-hand side is at least $\rho_{\min} a_n \rho_K^{(n-1)/2}\big(\frac{(\kappa-1)\ell_N}{2}\big)^{(n+1)/2} = 32u_N$ by the definition of ℓ_N and $w \ge \ell_N$. If $L > \rho_K$, it is at least $\rho_{\min} a_n \rho_K^n \ge 32u_N$ by the choice of N. In both cases (2) holds, and the theorem follows with C_1 determined by $u_N = O\big(\frac{n\log N}{N}\big)$ for $k=0$. □

4 Truncations: the corner-gap obstruction

Proof of Theorem 1.3. Write $e := \mathbf{1}/\sqrt{n}$, $\tau(x) := \langle x, e\rangle$, and $\tau_{\max} = \sqrt{n}$, attained at the corner $\mathbf{1}$. The region $G^* = \{x \in K : \tau(x) \ge \tau_{\max} - \eta_0\}$ with $\eta_0 := \frac{1}{2\sqrt{n}}$ is a simplex cap at the corner. It contains a ball of radius $r_0 = \frac{1}{4n}$: the point $z := \big(1-\frac{\eta_0}{2\sqrt{n}}\big)\mathbf{1}$ has $\tau(z) = \sqrt{n}\,\big(1-\frac{\eta_0}{2\sqrt{n}}\big) = \tau_{\max} - \frac{\eta_0}{2}$; each coordinate of z equals $1-\frac{\eta_0}{2\sqrt{n}}$, so $B(z,r) \subseteq [0,1]^n$ for $r \le \frac{\eta_0}{2\sqrt{n}} = \frac{1}{4n}$; and on $B(z,r)$ one has $\tau \ge \tau(z) - r \ge \tau_{\max} - \eta_0$ since $r \le \frac{\eta_0}{2}$. Hence $B\big(z, \frac{1}{4n}\big) \subseteq G^*$.

For $h \in [0,\eta_0]$ let $\Lambda(h) := \mu\big(\{x \in K :\ \tau(x) \ge \tau_{\max} - h\}\big)$ be the corner-cap mass. The cap $\{\tau \ge \tau_{\max} - h\}$ in the cube is the corner simplex $\{\sum_j (1-x_j) \le h\sqrt{n}\}$, of volume $\frac{(h\sqrt{n})^n}{n!}$ for $h\sqrt{n} \le 1$; thus

$$\Lambda(h) = \frac{n^{n/2}}{n!}\, h^n \qquad \text{for } 0 \le h \le \tfrac{1}{\sqrt{n}}. \tag{5}$$

Fix $\kappa, \sigma, \delta \ge \frac{3}{N}, \beta$ and set $\Lambda_0 := \frac{n^{n/2}}{n!}$, $h_N := \big(\frac{1}{\Lambda_0 N}\big)^{1/n}$ and $h^* := 4\kappa^2 h_N$, so $\Lambda(h_N) = \frac{1}{N}$ and $\Lambda(h^*) = \frac{(4\kappa^2)^n}{N}$. Consider the event

$$E := \Big\{\text{exactly one sample lies in the cap } \{\tau \ge \tau_{\max} - h^*\},\ \text{and it lies in } \{\tau \ge \tau_{\max} - h_N\}\Big\}.$$

As in the binomial computation of the sharpness section of [3],

$$\mathbb{P}(E) \;=\; N\,\Lambda(h_N)\big(1-\Lambda(h^*)\big)^{N-1} \;\ge\; N\cdot\frac{1}{N}\Big(1-\frac{(4\kappa^2)^n}{N}\Big)^{N-1} \;\ge\; c_3(n,\kappa) \;>\; 0$$

for all large N. Work on E and let X^* be the unique sample point in the h^*-cap, with depth $h_1 := \tau_{\max} - \tau(X^*) \le h_N$.

Consider the truncated slabs S_w with direction $u := e$, center $s := X^*$, and truncation set h^* (region G^*); every such slab contains X^*, so $\bar\mu_N(S_w) \ge \frac{1}{N} > 0$ for all $w > 0$. On the other hand, since $S_w \subseteq G^* \cap \{\tau \ge \tau(X^*) - \frac{w}{2}\} \subseteq \{\tau \ge \tau_{\max} - h_1 - \frac{w}{2}\}$ and $h_1 \le h_N = \frac{h^*}{4\kappa^2} \le \frac{h^*}{4}$, every slab of width $w \le h^*$ satisfies $S_w \subseteq \{\tau \ge \tau_{\max} - h^*\}$, the cap containing no sample other than X^* on the event E. Hence for all $w \le h^*/\kappa$,

$$\bar\mu_N(S_w) \;=\; \bar\mu_N(S_{\kappa w}) \;=\; \frac{1}{N} \;<\; \min\Big\{\delta,\ \frac{1+\sigma}{N}\Big\} \;=\; \min\Big\{\delta,\ (1+\sigma)\,\bar\mu_N(S_w)\Big\},$$

because $\delta \ge \frac{3}{N} > \frac{1}{N}$ and $\sigma > 0$. Thus the doubling inequality is violated at every scale $w \le h^*/\kappa$.

If $\ell \le h^*/\kappa^2$, the grid scale $w = \ell\kappa^i$ with $i = 0$ already lies in $(0, h^*/\kappa]$ (and $\ell < \beta$ for large N), so the UDC fails on E. Hence on E the UDC forces $\ell > h^*/\kappa^2 = 4h_N = c_2(n,\kappa)N^{-1/n}$. □

Remark 4.1. *The example uses the corner of the cube only for the explicit cap volume (5); any G whose level sections in some direction degenerate like h^{n-1} — equivalently, any truncation region with a conical point of its closure in the relevant direction, no matter how thick the region is globally — produces the same obstruction. Smoothness of K is irrelevant: the truncation hyperplanes create the corners. This is why the truncated exponent $\frac{1}{n}$ cannot be improved by regularity assumptions on the data support alone.*

5 Endpoint questions

Problem 5.1. *Determine the critical scaling at the logarithmic level: does the UDC for pure slabs on the ball hold with $\ell_N \asymp N^{-2/(n+1)}$ (no logarithm), or is a logarithmic factor necessary? The upper bound consumes $\log N$ through the VC bound of Lemma 2.2; the lower-bound construction of [3] produces only a single violating slab, and a union over $\asymp N^c$ essentially independent boundary caps should improve it to $\ell \geq c\,(\log N/N)^{2/(n+1)}$, matching the upper bound. We leave the details to future work.*

Problem 5.2. *Interpolate between the two regimes: for truncation regions whose boundary consists of smooth pieces meeting along edges of dimension j $(0 \leq j \leq n-1)$, determine the critical exponent of the scaling factor as a function of the lowest edge dimension present in the truncation family. The two theorems of this paper are the extreme cases: rolling-ball boundary (no edges) gives $\frac{2}{n+1}$, and a conical point ($j = 0$) gives $\frac{1}{n}$; we expect the level-section degeneracy at a j-dimensional edge to produce an intermediate exponent, monotone in j.*

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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